



Enhancing zietgebers

Investigating the effects of ageing on sleep and circadian rhythms.
<https://digit.in/sep21-25>



Life without temporal cues

This is an experiment on a man, living in isolation without any time cues.
<https://digit.in/sep21-26>

APPS TO TRACK CIRCADIAN RHYTHMS

Install these smartphone apps to keep track of your daily biorhythms. Note that while all these apps are free to install, the advanced functionalities require subscription services.



Rise: Sleep & Energy Tracker

Rise focuses on the most well studied and well understood aspect of tracking circadian rhythms, that is just getting a good night's sleep to match your natural melatonin window. You can track periods of insufficient sleep through the "sleep debt" calculated by the app. It also shows you peak energy levels.



Tempori

Tempori is a simple to use app with great watch integration, that requires minimal user data or inputs. It works by showing a suggested activity



Circadian

Circadian is the app that you need to use if you understand the theory and are an astronomy nerd. Alerts are a premium feature but you can set up stuff such as solar noon (which is not at 12:00 pm everyday) and alerts for periods of high UVA and UVB exposure.



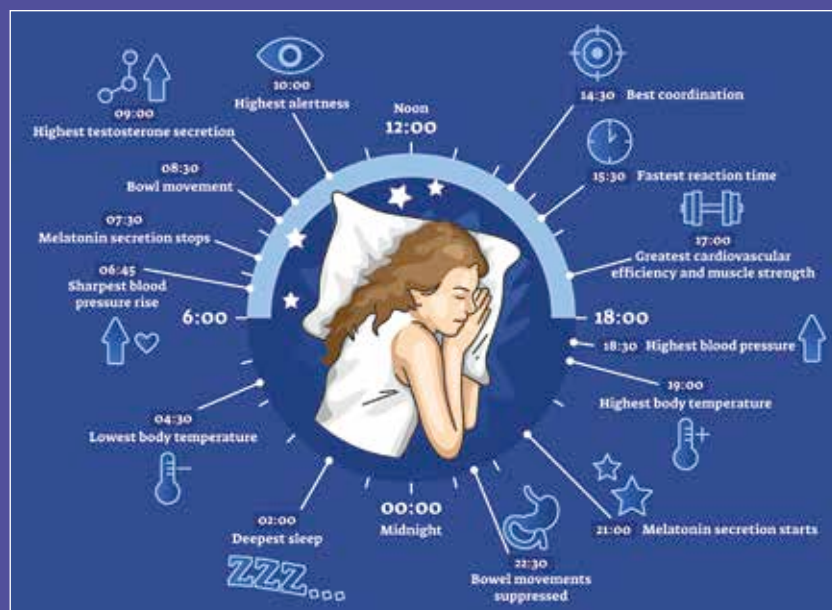
MyCircadianClock

This is a much more involving app as it requires you to track your eating habits, physical activity and rest periods manually through the app, and provides insights about your personal circadian rhythms. The data collected by the app is part of scientific research.

In the night, the melatonin rushes across the body from the pineal gland, signalling to all the cells that it is night time. This is how the SCN keeps the rest of the cells in the body synced with the actual day and night times. There are also clock cells distributed throughout the body that provide input to the local tissues. For example, the nutrients triggered by digestion can set the phase of oscillators in the tissues of livers, without this local entrainment passing on the information back to the SCN, which can independently

also keep these oscillators in phase. The function of the SCN is to keep this entire network of clocks across the body in phase with each other, and they can go topsy turvy in a number of different ways. The feeling of being jet lagged is all the different cells, and all the different clocks in the body being out of phase. Those who work in erratic or night shifts experience more acute consequences which only get worse the more the internal clock is out of sync with the external day and night cycle.

A complex series of inputs, oscillators, pacemakers and outputs together form the biological clocks in vertebrates. Every now and then, a novel component may be found in a fish or a bird, adding to our understanding. A lot of what we know about the entire molecular machinery for internal time-keeping synced to external cues, are from animals only, as it is very difficult to conduct these sorts of experiments on humans or open them up to see which hormones are being secreted when and where. For example, after a series of experiments on mutant fruitflies and rats, researchers uncovered the CLOCK (Circadian Locomotor Output Cycles Kaput) gene made the creatures have days that were either too long or too short, making them experience disrupted sleep patterns, and in the mice, mood swings. When the synchronisation is maintained, it is known as entrainment. Disruption of the synchronisation in humans is associated with sleep disruption, metabolic disorders, hypertension, arrhythmias and adverse effect on mental health. Light based therapies are being used to treat jet lag and delayed sleep phase syndrome, and researchers are actively investigating the role played by the zietgebers, particularly light and temperature, when it comes to keeping the body synced with the planet. **d**



The circadian rhythm of a normalised day